

An endpoint only a machine can buy from

Pragma.Vision

DEFINITION OF DONE — MET

GET /reading is gated with the x402 protocol on Cloudflare Workers. Two payments settled on Base Sepolia — one against a local server, one against the public endpoint — each returning the sensor reading and a verifiable transaction hash. *No account, no API key, no human in the loop. Testnet only; the tokens have no real value.*

DEFINITION OF DONE as written in the handbook

REQUIREMENT	STATUS	EVIDENCE
Worker route answers 402 with the payment-requirements payload	MET	Figures 2 and 5; docs/402-transcript.txt
Buyer script signs the payment authorization and retries	MET	Figures 4 and 6; buyer/pay.mjs
Verify and settle via the facilitator; transaction hash logged	MET	Two hashes below, both on the block explorer
Second call returns data plus a receipt	MET	status: 200 with reading and settlement object
curl transcript and explorer link in the README	MET	Figure 7 — committed and pushed

SETTLEMENTS independently verifiable

#	ENDPOINT	TRANSACTION
1 — 27 Jul	Local wrangler dev	0xe18db4768d05030511485080ad850270b49e8a00df7965470a27ae2b93f4d1f3
2 — 28 Jul	Public Worker on workers.dev	0xc5b68a953aa2ff89162a46c4a378d8c5fe35e3b86f77cf32dfbb4c2b403e92ff

CONFIGURATION as deployed

Network	eip155:84532 — Base Sepolia
Price	\$0.001 USDC per reading
Scheme	exact — EIP-3009 transferWithAuthorization
Facilitator	https://x402.org/facilitator
Buyer	Account 1 — funded with faucet USDC
Seller	Account 2 — empty, receives payment only

Buyer and seller are deliberately different addresses, so the explorer shows a genuine transfer between two parties rather than a self-payment.

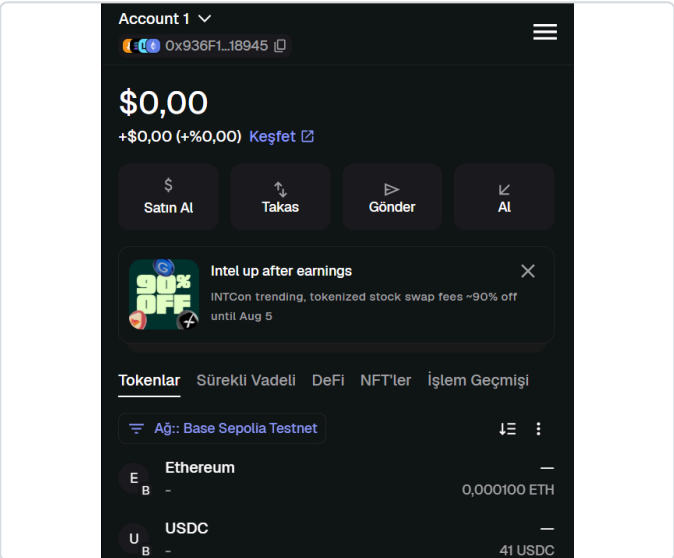


Fig. 1 — Buyer funded. 41 test USDC on Base Sepolia. Note the ETH balance: 0.0001. The payment still settled, which confirms the gasless property of the exact scheme in practice rather than in theory — the facilitator submits the transaction and pays the gas.

Fig. 2 — The closed door. An unpaid request returns 402 Payment Required. The base64 PAYMENT-REQUIRED header carries the terms: scheme, network, amount (1000 = \$0.001, since USDC has six decimals), the USDC contract, the seller address and a 300-second timeout. Everything a buyer needs, and nothing a human has to read.

Fig. 3 — Same URL, different customer. A browser sends `Accept: text/html` and receives a human-readable paywall instead of machine terms. One endpoint serves both audiences with no second integration — a small design detail with a large commercial implication.

local 27 July · deployed 28 July

Fig. 4 — Payment and delivery. The buyer decodes the terms, signs an EIP-3009 authorization, retries the same request with a payment header, and receives `status: 200` with the reading and a settlement object: `success: true`, the payer, the transaction hash and the network. The exchange is worth a tenth of a cent and involved no account, no card and no human decision.

Fig. 5 — Public endpoint gated. The deployed Worker on `workers.dev` returns 402 with the same terms. Server: `cloudflare` and the `CF-RAY` header confirm the request reached the edge rather than a local process. This is the address a stranger's agent will pay in Week 7.

```
{
  "buyer": "0x036f147d5489f a2236827bd5bc98c6b104718945",
  "status": 200,
  "body": {
    "device_id": "slm-sensor-01",
    "temperature_c": 22.49,
    "ts": "2026-07-20T15:21:00.571Z",
    "note": "TESTNET - no real value"
  }
},
  "settlement": {
    "success": true,
    "payer": "0x036f147d5489f a2236827bd5bc98c6b104718945",
    "transaction": "0xc5b68a953aa2ff89162a46c4a378db8c5fe35e3886f77cf32dfb4c2b403e92ff",
    "network": "eip155:84532"
  }
}
explorer: https://sepolia.basescan.org/tx/0xc5b68a953aa2ff89162a46c4a378db8c5fe35e3886f77cf32dfb4c2b403e92ff
```

Fig. 6 — Second settlement, public endpoint. A second, independent payment against the deployed Worker. Two settlements on two endpoints is the difference between a lucky run and a working mechanism.

```
akifk@Sorest MINGW64 ~/x402-iot-poc (main)
$ git add -A
git commit -m "feat(w2t2): x402-gated /reading with first Base Sepolia settlement"
git push
warning: in the working copy of 'wrangler.jsonc', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'docs/402-transcript.txt', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of 'docs/402-transcript.txt~', LF will be replaced
by CRLF the next time Git touches it
[main d463c0c] feat(w2t2): x402-gated /reading with first Base Sepolia settlement
7 files changed, 503 insertions(+), 24 deletions(-)
create mode 100644 buyer/pay.mjs
create mode 100644 docs/402-transcript.txt
create mode 100644 docs/402-transcript.txt~
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 20 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 6.76 KiB | 2.25 MiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/Soresta/x402-iot-poc.git
7e13f08..d463c0c main -> main
```

Fig. 7 — Committed. Code, buyer script and captured transcript pushed to the public MIT-licensed repository. Secrets stayed in a gitignored .env; a committed .env.example documents what a stranger must supply.

WHAT BROKE, AND WHAT IT COST

reported rather than smoothed over

- 1 **The dev server kept dying.** Pasting curl commands into the terminal running wrangler dev triggered its single-key shortcuts — t opened a tunnel, x quit. Two hours were lost to a server that appeared to crash on its own. Fix: never type in that window; run every test from a second terminal.
- 2 **An empty evidence file.** Capturing the transcript with curl -s silenced errors as well as progress output, so a failed request wrote a file containing only a timestamp. Fix: -sS keeps errors visible. A capture command that can fail silently is worse than none.
- 3 **Invalid private key.** The buyer crashed inside the signing library because the key in .env was not 66 characters of hex. Diagnosed without printing the key by checking its length with grep and awk — the habit matters more here than the testnet value does.
- 4 **Port drift.** Orphaned processes held 8787, so restarts moved to 8788 while tests still pointed at the old port. Fix: read the port from the “Ready on...” line every time rather than assuming it.

FINDINGS WORTH PUBLISHING

neither is documented anywhere I found

- 1 **Cloudflare Workers needs lazy middleware construction.** Building the x402 payment middleware at module scope fails, because Workers restricts cryptographic operations in the global scope. It has to be constructed inside the request handler on first use. No official example mentions this, and the resulting failure is not self-explanatory.
- 2 **Two incompatible package generations are in circulation.** The legacy family uses network: "base-sepolia"; the current scoped family uses CAIP-2 identifiers such as eip155:84532. Vendor documentation is split between them, and combining a server from one generation with a client from the other fails without a useful error. This is the most likely thing to block a newcomer, and it is not flagged as a warning anywhere.

Both are now recorded in the repository README. The second is the strongest candidate for a first genuinely useful community post: it costs other people real time, and reporting it requires no self-promotion.

WHAT THIS DOES NOT YET DO

stated before anyone asks

A replayed payment proof is not yet rejected — the same signature would currently pay twice. There is no spending mandate with caps and an expiry, no rate limiting and no dispute handling. The sensor is simulated. All four are the substance of Week 3 rather than oversights, and the README lists them in the same words.

Assessment. The settlement layer is not the hard part. Gating an endpoint and settling a payment took roughly twenty lines of middleware and one evening, and most of the time actually spent went to tooling friction rather than to protocol difficulty. That is the finding worth carrying into the use-case decision: what remains unbuilt above settlement — authorization, replay protection, trust — is where the engineering and the commercial risk both live.